



Cross-sectional Imaging of the Duodenum: Spectrum of Disease

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Although upper gastrointestinal endoscopy plays a well-established role in duodenal evaluation, it allows direct visualization of only the proximal lumen. Cross-sectional imaging modalities (especially CT and MRI) have been increasingly effective in staging duodenal lesions, some of which are incidentally detected and beyond the limits of endoscopic evaluation. The online presentation comprises an extensive and illustrative review of the clinical and radiologic aspects of multiple duodenal conditions, which are classified into five main categories.

Infectious and inflammatory causes are tuberculosis, peptic ulcer disease, pancreatic pseudocyst in the duodenal wall, paraduodenal or groove pancreatitis, Crohn disease, cholecystoduodenal fistula (Bouveret syndrome), and Zollinger-Ellison syndrome. Malignant masses are adenocarcinoma, lymphoma, neuroendocrine tumor, and leiomyosarcoma. Benign masses are hamartomatous polyp, lipoma, gastrointestinal stromal tumor, villous adenoma, schwannoma, and gangliocytic paraganglioma. Traumatic and vascular causes are hematoma, aortoenteric fistula, superior mesenteric artery syndrome, and Henoch-Schönlein purpura. Anatomic causes are bulging papilla, heterotopic pancreas, intussusception, duplication cyst, and Lemmel syndrome.

Correlation of cross-sectional imaging findings with endoscopic findings is included in the online presentation when available. Understanding the anatomic features of each duodenal portion can also aid in diagnosis. Examples include peptic ulcer disease in the first portion of the duodenum due to the proximity to gastric acid (Fig 1), pancreatic pseudocyst in the second portion of the duodenum in proximity to the pancreatic parenchyma (Fig 2), and Crohn disease involving the fourth portion of the duodenum in conjunction with a proximal jejunal stricture.

TEACHING POINTS

- The abdomen is one of the most common sites of extrapulmonary tuberculosis, but duodenal involvement accounts for fewer than 2% of cases. This can be partially explained by the inhibitory influence of gastric acid on mycobacteria with a rapid transit time through the duodenum and a relative paucity of lymphoid tissue in this segment as compared to the rest of the gastrointestinal tract.
- The duodenum is the most common site of intramural hematoma of the gastrointestinal tract. As blunt abdominal trauma is the main cause, the duodenal segments most likely to develop mural hematoma related to abrupt avulsion are the third horizontal portion and the duodenojejunal flexure, owing to the lack of mesentery and to fixation by the Treitz ligament, respectively.
- Despite their size, gastrointestinal stromal tumors (GISTs) rarely cause duodenal obstruction or obstructive jaundice, which can be attributed to a predominantly extraluminal growth pattern. At imaging, a malignant GIST appears as a large well-circumscribed mass with a heterogeneously enhancing pattern and may contain necrotic areas, ulceration, and a central cavity with or without an air-fluid level.

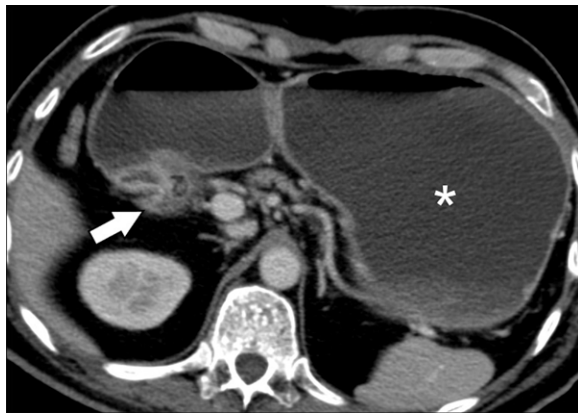


Figure 1. Penetrating peptic duodenal ulcer in a 54-year-old man with acute epigastric pain and hematemesis. Axial CT image reveals inflammatory changes involving the duodenal bulb, including hyperenhancing mucosa, a luminal outpouching with regional fat stranding (arrow), and gastric dilatation (*).



Figure 2. Pancreatic pseudocyst in the duodenal wall in a 42-year-old man with a recent history of acute pancreatitis. Coronal CT image depicts a fluid collection along the duodenal wall (dashed line) and gastric distention (*). Sampling of the fluid collection revealed a high amylase concentration (1200 U/L).

Careful imaging assessment of the duodenum that combines clinical and epidemiologic aspects with key imaging features facilitates rapid and accurate diagnosis and treatment.

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